RetinaCare Research Screening Report

RESEARCH PROTOTYPE - NOT A CLINICAL DOCUMENT. For investigational screening only; not a diagnosis. A licensed ophthalmologist must examine the patient.

1. Identification

Patient code: E2E-001
Patient name: Integration Test Patient
Examination: EXAM-001
Eye: OD (right)
Examination date: 2026-10-28T07:50:38Z
Source image: OD-1.png (640 x 640 px)
Report generated: 2026-10-28 07:56:50Z

2. Status

COMPLETE - Stage I, Stage II (5 masks) and Stage III fold 0 (single-fold technical test)

3. Results

Outcome: Single-fold technical severity grade 3 (research only)
SINGLE-FOLD TECHNICAL TEST (LG-DRG fold 0) - RESEARCH ONLY. Not a clinical grade; five-fold ensemble, licensed-image acceptance and clinical validation are pending.
Stage I screening gate: score 0.9983 (not a calibrated probability), threshold 0.2392, raw logit 6.386, route: continue cascade.
Stage II microaneurysms (MA): 58 foreground pixels of 640 x 640 (0.014% of the frame).
Stage II hemorrhages (HE): 1574 foreground pixels of 640 x 640 (0.384% of the frame).
Stage II exudates (EX): 8997 foreground pixels of 640 x 640 (2.197% of the frame).
Stage II cotton-wool spots (CWS): 400 foreground pixels of 640 x 640 (0.098% of the frame).
Stage II vessels (SwinHRUNet++): 47521 foreground pixels of 640 x 640 (11.602% of the frame).
Stage III LG-DRG (single-fold technical test (fold 0)): technical severity grade 3; gradable probability 1.000.

4. Provenance

Stage I model: hierarchiretina-stage1-convnextv2l-768 sha256 9ad51e1f428dc59ab49f1f49e8f68a84cd23d8fd4c2c461b45b06bcc726880ac (CPU)
Stage II microaneurysms (MA) model sha256 ac874bee868bbd7109d1358deaa35ab59f38d7664a4ea7eec22e00c8be79aac5 (CPU)
Stage II hemorrhages (HE) model sha256 16f767a8deb3751962d00abfaa5aca27fb40b43c64a77108c78610111260b966 (CPU)
Stage II exudates (EX) model sha256 2aa92099f2333e3d1afe862a618a8436023bc03f14b5a8f62ea67debf2612207 (CPU)
Stage II cotton-wool spots (CWS) model sha256 6c928a75b28884ca19a58fa11daa6020868af57081b07b347961ce7b4c967882 (CPU)
Stage II vessels (SwinHRUNet++) model sha256 cc7ef0bb23732c7a1c89954244df9b6a5523f516c3818821811c667d2d5d76e0 (CPU)
Stage III model sha256 31f20d62f972a0a8f9119162c8a30b7e2a96509933f0a9bc9a3845b3a6129cb7 (CPU)
Source image sha256: 32d756fc923262daa5296c6852c1569e01394b37db57e72385785fca9017c899; identity checks (patient, examination, eye, image, frame size): passed.
Measured run time 73.5 s, peak memory 2951 MiB.

5. Limitations

- Research prototype. No clinical validation; not a medical device.

- Stage III uses fold 0 of five folds only. Folds 1-4 and the ensemble are not implemented.
- No licensed or clinically validated image set has been used for acceptance.
- Only the CPU execution provider has been exercised; CUDA is not verified.
- Model weights are CC BY-NC 4.0 (non-commercial use only).
- This PDF contains text only: segmentation masks and the source image are not embedded.
- Explainability (heatmaps) is not available and none is shown.